CHENBEI LU

100 Fairview Sq, Ithaca, NY 14850 ⋄ chenbei.lu@cornell.edu ⋄ www.luchenbei.com

Updated September 2026

RESEARCH PROFILE

My research develops data-efficient, scalable, and reliable reinforcement learning and control by exploiting system structure and imperfect predictions. Motivated by power-system operation, I study sample complexity, risk, and stability, and connect these foundations to energy-storage control and the planning and operation of power and AI computing infrastructure.

ACADEMIC AND RESEARCH APPOINTMENTS

Cornell University AI for Science Institute — *Ithaca, NY, USA*
Eric and Wendy Schmidt AI in Science Postdoctoral Fellow — Sep. 2025–Present
Host: Prof. Fengqi You

EDUCATION

Tsinghua University Institute of Interdisciplinary Information Sciences — *Beijing, CN*
Ph.D. in Computer Science — Sep. 2020–Jun. 2025
Advisors: Prof. Chenye Wu and Prof. Ran Duan

California Institute of Technology Computing & Mathematical Sciences — *Pasadena, CA, USA*
Visiting Student Researcher — Aug. 2023–Feb. 2024
Advisor: Prof. Adam Wierman

Huazhong University of Science & **Technology** School of Software Engineering — *Wuhan, CN*
B.E. in Computer Software Engineering — Sep. 2016–Jun. 2020
GPA 3.97/4.00 (Rank 1/180), Outstanding Graduate, Outstanding Bachelor's Thesis Award

PUBLICATIONS

$*$ indicates equal contribution.

Peer-Reviewed Publications: Published and Accepted

Reinforcement Learning Theory and Learning-Based Control

1. **Chenbei Lu** and Hongyu Yi. "Learning Chance-Constrained MDPs with Bellman Distributional Certificates." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.

2. Hongyu Yi, **Chenbei Lu**, and Jing Yu. "Achieving $\widetilde{\mathcal{O}}(1/\varepsilon)$ Sample Complexity for Bilinear Systems Identification under Bounded Noises." *IEEE Control Systems Letters*, 2026.

3. **Chenbei Lu**, Zaiwei Chen, Tongxin Li, Chenye Wu, and Adam Wierman. "Reinforcement Learning with Imperfect Transition Predictions: A Bellman–Jensen Approach." *Conference on Neural Information Processing Systems (NeurIPS)*, 2025, **Spotlight**.
Extended journal version: "Reinforcement Learning with Transition Predictions: A Bellman Theory" (manuscript in preparation).

4. **Chenbei Lu**, Laixi Shi, Zaiwei Chen, Chenye Wu, and Adam Wierman. "Overcoming the Curse of Dimensionality in Reinforcement Learning Through Approximate Factorization." *International Conference on Machine Learning (ICML)*, 2025.

5. Hongyu Yi*, **Chenbei Lu***, and Chenye Wu. "On the Optimal Deterministic Policy Learning in Chance-Constrained Markov Decision Processes." *IEEE Control Systems Letters*, 2025.

Power and Energy Systems: Optimization, Control, and Markets

6. **Chenbei Lu** and Fengqi You. "Online Neural Fusion of Deep Reinforcement Learning and Model Predictive Control for Comfort-Aware Greenhouse Energy Optimization." *Applied Energy*, 426:128621, 2026.

7. **Chenbei Lu**, Jinhao Liang, Hongyu Yi, and Chenye Wu. "Cost-Effective Closed-Loop Bilevel Robust Optimization for Joint Chance-Constrained Economic Dispatch." *ACM International Conference on Future and Sustainable Energy Systems (ACM e-Energy)*, 2025.

8. **Chenbei Lu**, Hongyu Yi, Jiahao Zhang, and Chenye Wu. "Self-Improving Online Storage Control for Stable Wind Power Commitment." *IEEE Transactions on Smart Grid*, 2024.

9. Gaoyuan Xu, Jian Shi, Jiaman Wu, **Chenbei Lu**, Chenye Wu, Dan Wang, and Zhu Han. "An Optimal Solutions-Guided Deep Reinforcement Learning Approach for Online Energy Storage Control." *Applied Energy*, 2024.

10. Jinhao Liang, Wenqian Jiang, **Chenbei Lu**, and Chenye Wu. "Joint Chance-Constrained Unit Commitment: Statistically Feasible Robust Optimization with Learning-to-Optimize Acceleration." *IEEE Transactions on Power Systems*, 2024.

11. **Chenbei Lu**, Nan Gu, Wenqian Jiang, and Chenye Wu. "Sample-Adaptive Robust Economic Dispatch With Statistically Feasible Guarantees." *IEEE Transactions on Power Systems*, 2023.

12. Wenqian Jiang, **Chenbei Lu**, and Chenye Wu. "Robust Scheduling of Thermostatically Controlled Loads with Statistically Feasible Guarantees." *IEEE Transactions on Smart Grid*, 2023.

13. **Chenbei Lu**, Jingshi Cui, Haoxiang Wang, Hongyu Yi, and Chenye Wu. "Privacy-Preserving User Energy Consumption Profiling: From Theory to Application." *IEEE Transactions on Smart Grid*, 2023.

14. **Chenbei Lu**, Jinhao Liang, Nan Gu, Haoxiang Wang, and Chenye Wu. "Manipulation-Proof Virtual Bidding Mechanism Design." *IEEE Transactions on Energy Markets, Policy and Regulation*, 2023.

15. Jiaman Wu, **Chenbei Lu**, Chenye Wu, Jian Shi, Marta C. Gonzalez, Dan Wang, and Zhu Han. "A Cluster-Based Appliance-Level-of-Use Demand Response Program Design." *Applied Energy*, 362:123003, 2024.

16. **Chenbei Lu**, Wenqian Jiang, and Chenye Wu. "Effective End-to-End Learning Framework for Economic Dispatch." *IEEE Transactions on Network Science and Engineering*, 9(4): 2673–2683, 2022.

17. Jiaman Wu, **Chenbei Lu**, and Chenye Wu. "Learning-Aided Framework for Storage Control Facing Renewable Energy." *IEEE Systems Journal*, 2022.

18. **Chenbei Lu**, Jiaman Wu, and Chenye Wu. "Privacy-Preserving Decentralized Price Coordination for EV Charging Stations." *Electric Power Systems Research*, 212:108355, 2022.

19. **Chenbei Lu**, Jiaman Wu, Jingshi Cui, Yanyan Xu, Chenye Wu, and Marta C. Gonzalez. "Deadline-Differentiated Dynamic EV Charging Price Menu Design." *IEEE Transactions on Smart Grid*, 2022.

20. **Chenbei Lu**, Zhiqi Wang, and Chenye Wu. "Storage-Aided Service Surcharge Design for EV Charging Stations." *IEEE Conference on Decision and Control (CDC)*, 2021.

Preprints and Manuscripts Under Review

M1. **Chenbei Lu** and Fengqi You. "AI Data-Center Growth Exposes Regional Electricity-Supply Headroom Constraints in the United States." Revise and resubmit at *Nature Communications*.

M2. **Chenbei Lu** and Zaiwei Chen. "The Statistical Value of Lyapunov Stability in Reinforcement Learning." Under review at the *International Conference on Learning Representations (ICLR)*, 2027.

M3. **Chenbei Lu** and Laixi Shi. "Bellman-Convex Reinforcement Learning in Continuous Markov Decision Processes." Under review at the *International Conference on Learning Representations (ICLR)*, 2027.

TEACHING AND MENTORING

Guest Lectures

- **Deep Learning**, Cornell University, Spring 2026. Designed and delivered two one-hour guest lectures on reinforcement learning.
- **AI for Science**, Cornell University, Fall 2025 and Fall 2026. Delivered introductory reinforcement-learning guest lectures in two course offerings.
- **AI for Energy Systems**, Cornell University, Fall 2025. Designed and delivered a one-hour guest lecture on energy-storage systems and control.

Teaching Assistantships

- **AI Research Practice**, Tsinghua University, Fall 2022. Teaching assistant for a 30-student Yao Class undergraduate course; tracked approximately ten student research projects and provided technical feedback.
- **Combinatorial Mathematics**, Tsinghua University, Fall 2021. Teaching assistant for an approximately 90-student graduate course; graded weekly problem sets and held office hours.

Student Co-Mentoring

- **Hongyu Yi** (Sep. 2022–Present): chance-constrained reinforcement learning and online energy-storage control. Joint work includes papers in *IEEE Control Systems Letters* (2025) and NeurIPS (2026).
- **Jiajun Han** (Jun. 2024–Jul. 2025): up-to-congestion bidding under market risk and credit exposure.
- **Jinhao Liang** (Sep. 2022–Dec. 2023): robust unit commitment with sample adaptivity and learning-to-optimize acceleration. Co-authored work in *IEEE Transactions on Power Systems* (2024) and at ACM e-Energy (2025).

SELECTED TALKS AND PRESENTATIONS

1. "**Reliability-First Reinforcement Learning for Coordinated Voltage and Storage Control in Renewable-Rich Grids**." National Academies AI in the Electricity System Workshop, Washington, DC, USA, October 2026. *Scheduled poster presentation.*
2. "**Reinforcement Learning with Imperfect Transition Predictions: A Bellman–Jensen Approach**." Conference on Neural Information Processing Systems (NeurIPS), Spotlight Presentation, San Diego, USA, December 2025.
3. "**Overcoming the Curse of Dimensionality in Reinforcement Learning Through Approximate Factorization**." International Conference on Machine Learning (ICML), Vancouver, Canada, July 2025. *Conference presentation.*
4. "**Sample-Adaptive Robust Economic Dispatch With Statistically Feasible Guarantees**." IEEE Power and Energy Society General Meeting 2024, Seattle, USA, July 2024. *Conference presentation.*
5. "**Data-driven Power System Operation under Uncertainty: from Offline to Online Perspective**." National Outstanding Graduate Student Forum, The Chinese University of Hong Kong, Shenzhen, April 2024.
6. "**Sample Efficient Reinforcement Learning via Kernel Graph Decomposition**." Rigorous Systems Research Group Seminar, Caltech, Pasadena, USA, February 2024. *Invited seminar*; hosted by Prof. Adam Wierman.
7. "**Sample Efficient Reinforcement Learning via Kernel Graph Decomposition**." The Chinese University of Hong Kong, Shenzhen, November 2023. *Invited seminar*; hosted by Prof. Tongxin Li.
8. "**Storage-aided Service Surcharge Design for EV Charging Stations**." 60th IEEE Conference on Decision and Control, Virtual, December 2021. *Conference presentation.*

PROFESSIONAL SERVICE

Co-organizer: Cornell AI for Sustainability Seminar Series, 2026.

TPC Member: IEEE SmartGridComm, 2024; ACM e-Energy Poster Session, 2026.

Journal Reviewer: *IEEE Transactions on Pattern Analysis and Machine Intelligence*; *IEEE Transactions on Automatic Control*; *IEEE Transactions on Smart Grid*; *IEEE Transactions on Power Systems*; *IEEE Transactions on Sustainable Energy*; *IEEE Transactions on Transportation Electrification*; *IEEE Transactions on Industry Applications*; *IEEE Systems Journal*; *International Journal of Electrical Power & Energy Systems*; *Electric Power Systems Research*; *iScience*; and *Scientific Reports*.

Conference Reviewer: NeurIPS, ICLR, AAAI, CDC, ACC, PSCC, IEEE SmartGridComm, and ACM e-Energy.

INDUSTRIAL RESEARCH EXPERIENCE

Didi Chuxing Technology Co. Department of Data Science and Intelligence *Beijing, CN*
Research Intern Jan. 2021–Nov. 2021
Topics: Improving the Efficiency of Transportation Systems through Carpooling Design

SELECTED HONORS

- Eric & Wendy Schmidt AI in Science Postdoctoral Fellowship Sep. 2025
- IIIS Outstanding Graduate Award, Tsinghua University Jun. 2025
- Toyota Outstanding Scholarship, Tsinghua University Oct. 2024
- Outstanding Presentation Award, National Outstanding Graduate Student Forum Apr. 2024
- Ubiquant Outstanding Scholarship, Tsinghua University Oct. 2023
- National Scholarship, Huazhong University of Science & Technology Oct. 2017